

TOPICAL REVIEW

Gallium oxide Schottky diode defects and control

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TOPICAL REVIEW

Gallium oxide Schottky diode defects and control

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Abstract

With technological leaps in fields such as power electronics, energy, communications, and national defense, Due to the limitations of physical conditions, traditional silicon-based semiconductors are increasingly unable to meet the performance requirements of these applications. Therefore, gallium oxide ($\beta\text{-Ga}_2\text{O}_3$) has emerged as a highly promising alternative material. Gallium oxide (particularly $\beta\text{-Ga}_2\text{O}_3$), as a representative of fourth generation ultra wide bandgap semiconductor materials, holds strategic significance in current technological development for its device research. The high-temperature resistance of gallium oxide (with a junction temperature up to 200 °C) makes it irreplaceable in extreme environments such as aerospace and nuclear industries. The theoretical breakdown voltage of gallium oxide devices exceeds 10 kV, more than twice that of silicon carbide. As the fundamental building block of gallium oxide devices, the identification and control of defects in gallium oxide Schottky diodes are of paramount importance. This paper analyses the defects of gallium oxide Schottky diodes and summarizes how to control these defects, hoping to be a guide in subsequent research.

1. Introduction

Silicon (Si) has long served as the fundamental material for semiconductor devices, yet traditional silicon-based semiconductors increasingly fail to meet the performance requirements of new electronic products [1–4]. Consequently, ultra-wide bandgap (UWBG) semiconductors are being extensively researched to overcome these limitations and deliver superior performance. GaN, silicon carbide (SiC), and $\beta\text{-Ga}_2\text{O}_3$ have gradually emerged as promising alternatives, with their field effect transistor (FET) and diodes already appearing on the market [5, 6]. The properties of $\beta\text{-Ga}_2\text{O}_3$ are particularly noteworthy. $\beta\text{-Ga}_2\text{O}_3$ is a wide-bandgap semiconductor exhibiting α , β , γ , δ , and ε phases [7–11], as shown in figure 1. $\beta\text{-Ga}_2\text{O}_3$ is the most chemically and thermally stable phase among the five. It exhibits excellent material properties, including a wide bandgap of approximately 4.8 eV, a high breakdown electric field of about 8 MV cm⁻¹, and an impressive Baliga's figure of merit (3444). Compared to other materials, it offers advantages of ultra high voltage, low cost, and strong radiation resistance. Bulk crystal growth methods for $\beta\text{-Ga}_2\text{O}_3$, such as the Czochralski (CK) method and the floating zone method [12–14], are relatively inexpensive and straightforward. $\beta\text{-Ga}_2\text{O}_3$ can withstand radiation doses five times that of GaN and three times that of SiC. Therefore, $\beta\text{-Ga}_2\text{O}_3$ is considered a promising candidate for next-generation power electronic devices. Over the past decade, $\beta\text{-Ga}_2\text{O}_3$ based power devices, including FET and Schottky barrier diodes (SBD), have been extensively studied. Crystal quality, process optimization, and device structure engineering are critical for achieving satisfactory performance in $\beta\text{-Ga}_2\text{O}_3$ devices.

This paper reviews defects in $\beta\text{-Ga}_2\text{O}_3$ Schottky diodes and methods for their mitigation. The specific defects inherent in $\beta\text{-Ga}_2\text{O}_3$ itself and those introduced during the fabrication of gallium oxide Schottky diodes, along with defects induced in the Schottky diodes by irradiation, are illustrated in figure 2. Subsequently, it discusses methods to reduce the impact of these defects on Schottky diodes. Finally, the paper concludes with a summary of the entire content.

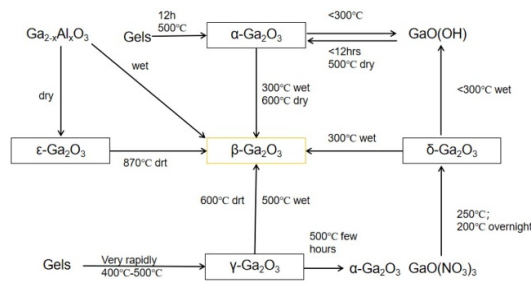


Figure 1. Conditions for the phase transitions of the four forms of gallium oxide.

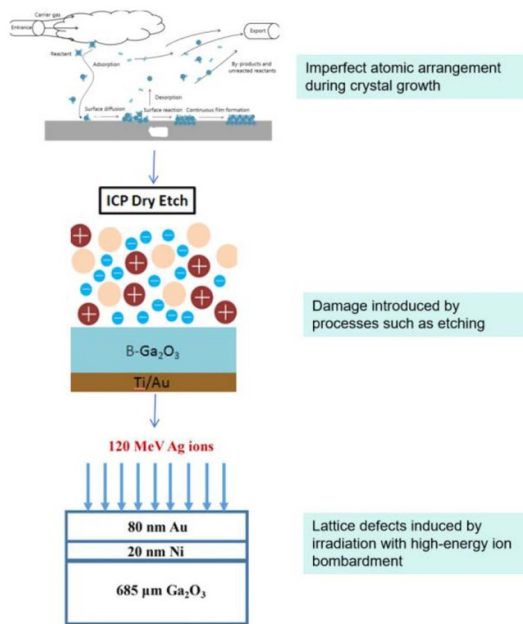


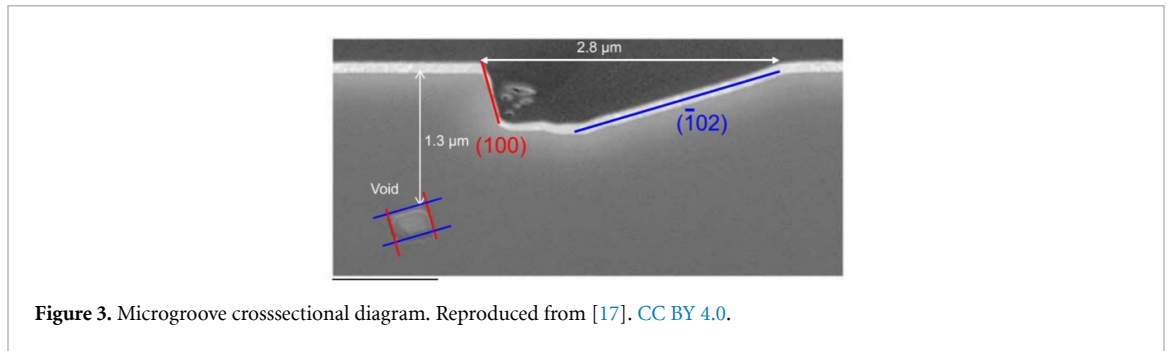
Figure 2. The lifecycle of Schottky diodes from fabrication to irradiation. Partly reprinted from [15], with the permission of AIP publishing. Partly reprinted from [16], with the permission of AIP publishing.

2. Defects in gallium oxide power devices and their effects

Gallium oxide devices hold broad application prospects and significant development potential in fields such as aerospace, specialized power supplies, new energy vehicles, and high power fast charging. Among these, gallium oxide Schottky diodes serve as the core components of gallium oxide power devices. Therefore, precise characterization, mechanism elucidation, and effective control of their critical defects are of paramount importance. This paper summarizes the typical defects present in Gallium oxide Schottky diodes across three major stages: intrinsic defects arising during gallium oxide material growth, interface and structural defects introduced during device fabrication processes, and radiation-induced damage and reliability degradation issues under irradiation environments.

2.1. Gallium oxide defects and their effects

It is well known that defects significantly affect the electrical properties of materials and devices. Therefore, the identification and thorough characterization of defects in β -Ga₂O₃ single crystals is crucial. These defects often serve as conduits for reverse leakage current, leading to a reduction in the diode's breakdown voltage. It is crucial to investigate their impact on gallium oxide diodes. Kasu's group [17] discovered groove like hole defects in β -Ga₂O₃ single crystals, and subsequently observed the same phenomenon on the epitaxial layer of β -Ga₂O₃ diodes. Both surfaces exhibited microgrooves, as shown in figure 3. The presence of microgrooves increases reverse leakage current, and the concentration of electric fields causes the diode to break down earlier. In addition to microgrooves, Labed and Sdoeung discovered in 2023 that linear defects also constitute a primary leakage current pathway in β -Ga₂O₃ diodes, particularly at low temperatures where dislocation defects dominate [18, 19]. Microgrooves and

**Table 1.** Defect level positions and concentrations for different growth methods.

Growth pattern	Trap energy level position from the bottom of the conduction zone (eV)	Trap concentration (cm^{-3})	References
Ionophore-assisted epitaxial growth of molecular beams	0.98, 2.00, 3.25, 4.37	1×10^{16} (2, 3.25, 4.37), 1.6×10^{15} (0.98)	[20]
Edge-defined thin film feedthrough growth	0.62, 0.82, 1.00, 2.16, 4.40	4.7×10^{14} (0.62), 3.6×10^{16} (0.82), 3.7×10^{15} (1), 1.0×10^{15} (2.16), 1.5×10^{16} (4.4)	[21]
Tilapha growth	0.55, 0.74, 1.04	1×10^{15} (0.55), 4×10^{16} (0.74, 1.04)	[22]
Straight pull method	0.6, 0.8, 1.1	1.1×10^{14} (0.6), 3.9×10^{16} (0.8), 8.9×10^{15} (1.1)	[23]

linear defects can be reduced or eliminated through novel growth methods or pretreatment, with specific optimization approaches discussed later in this paper.

Linear defects and microgrooves constitute geometric trench like structural defects on the surface, while deep level defects and shallow level defects represent electronic structural defects introducing intermediate energy states within the bandgap. Both deep and shallow defect states significantly impact device performance parameters through trap effects. Shallow traps, situated near the conduction or valence bands, readily capture and release charge carriers, affecting response speed and transient current. Deep traps, situated near the center of the bandgap, make it difficult for carriers to escape once trapped, leading to carrier depletion and leakage current. Therefore, understanding the existence of deep level and shallow level defect states across the entire β -Ga₂O₃ bandgap of epitaxial materials is crucial for the future development of reliable and high performance gallium oxide diodes. Different growth methods for gallium oxide result in varying trap level positions and concentrations, as shown in table 1 [20–24]. Adjusting trap level positions or reducing trap concentrations can optimize their effects on diodes or other devices.

2.2. Devices defects in making gallium oxide diodes

Etching and polishing processes serve as the core structural shaping and performance tuning steps in the fabrication of β -Ga₂O₃ power devices. Their quality directly determines whether the devices can leverage the intrinsic advantages of the β -Ga₂O₃ material. Polishing is a critical ‘post-processing step’ in gallium oxide device fabrication, fundamentally eliminating issues introduced by prior processes such as crystal cutting and mechanical grinding by improving interface quality. Perrier *et al* observed (100)-cracks and dislocations along the [100] direction via scanning electron microscopy [24], as shown in figure 4. This defect is presumed to have been generated by collisions between slurry accumulated during chemical mechanical polishing and the surface. Therefore, it is necessary to or eliminate these types of critical defects. β -Ga₂O₃ exhibits extreme chemical inertness toward most acid and alkali, resulting in extremely slow wet etching rates and poor selectivity [25–27]. Consequently, dry etching must be relied upon for structural processing. Ion bombardment during dry etching readily induces lattice dislocations, generating defects such as oxygen vacancies and dangling bonds [28–30]. Consequently, the selection of Etching gases is critical. The etching process directly determines the feasibility of forming the three dimensional structures essential for power devices in GaO, including vertical steps, trenches, and source/drain grooves. β -Ga₂O₃ single crystals exhibit pronounced lattice anisotropy. If the etching process is not aligned with the crystal orientation characteristics, it will result in unevenly inclined trench sidewalls and jagged defects at the bottom, directly disrupting the continuity of the current channel.

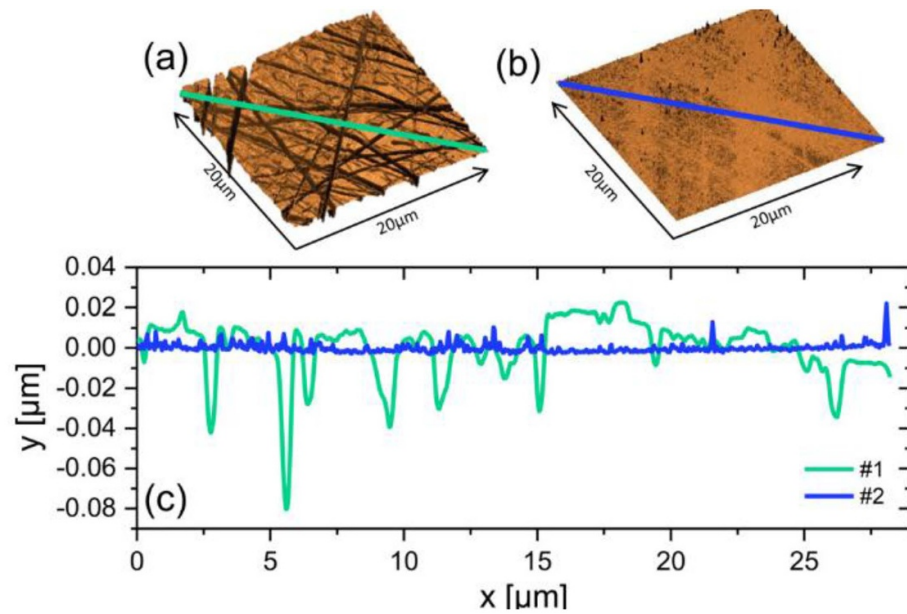


Figure 4. Example of AFM images with an area of 400 lm^2 near a Schottky contact of (a) #1 and (b) #2. (c) Cross profiles of the surface of #1 and #2 extracted from AFM images recorded along lines shown in (a) and (b). Reprinted from [24], with the permission of AIP publishing.

2.3. Defects induced in gallium oxide power devices by irradiation

Power devices fabricated from $\beta\text{-Ga}_2\text{O}_3$ exhibit enhanced radiation resistance [31–33]. These distinctive physical properties make $\beta\text{-Ga}_2\text{O}_3$ an ideal candidate material for aerospace applications. In space environments, the harsh radiation exposure can cause $\beta\text{-Ga}_2\text{O}_3$ power devices to fail. Since diodes constitute their fundamental structural components, investigating the irradiation effects on $\beta\text{-Ga}_2\text{O}_3$ diodes is of critical importance.

This section primarily analyzes the effects of neutron, proton, electron, and heavy ion irradiation on gallium oxide Schottky diodes. Neutron irradiation primarily interacts with the material through nuclear recoil, with ionization effects being negligible. Displacement damage serves as the primary mechanism, introducing intrinsic defects such as gallium vacancies and oxygen vacancies within the device. Proton irradiation exhibits both significant ionization effects and displacement damage, forming mixed shallow-level and reverse leakage current in devices is primarily dominated by Pool–Frenkel emission (PFE). Electron irradiation interacts with the material mainly through ionization energy loss, with negligible displacement damage. It primarily affects the oxide layer, interface regions, and shallow-level traps. Heavy ions (e.g., Ag, Ta, Kr, Ge) possess extremely high linear energy transfer values and highly localized energy deposition, forming continuous damage latent tracks within the material while inducing strong ionization effects. Building upon this, the study further analyzes the patterns of irradiation effects on key electrical parameters of diodes.

2.3.1. Neutron irradiation

Under high-flux neutron irradiation, oxygen atoms in the $\beta\text{-Ga}_2\text{O}_3$ layer undergo nuclear reactions that displace them from their original positions, forming a large number of oxygen vacancies [34]. These vacancies act as deep-level defects, compensating for carriers in the material. Simultaneously, neutron bombardment severely disrupts the $\beta\text{-Ga}_2\text{O}_3$ lattice, inducing displacement damage. These amorphous or polycrystalline regions reduce carrier mobility and concentration, ultimately leading to significant degradation in device performance.

Liu and Fu conducted irradiation tests on $\beta\text{-Ga}_2\text{O}_3$ diodes using 1 MeV and 14 MeV sources, respectively, as shown in figure 5 [35, 36]. Following 1 MeV neutron irradiation, the forward current density (J_F) decreased by 20%, the reverse current density (J_R) decreased by 75%, and the breakdown voltage increased. This increase in breakdown voltage resulted from a decrease in carrier concentration in the $\beta\text{-Ga}_2\text{O}_3$ drift layer, leading to a widening of the depletion layer. Concurrently, the density of interface states increased. This indicates that the degradation in diode performance is dominated by bulk trap concentration rather than interface trap concentration. Both reverse leakage current density and forward leakage current density increased under 14 MeV neutron irradiation, indicating that different kinetic

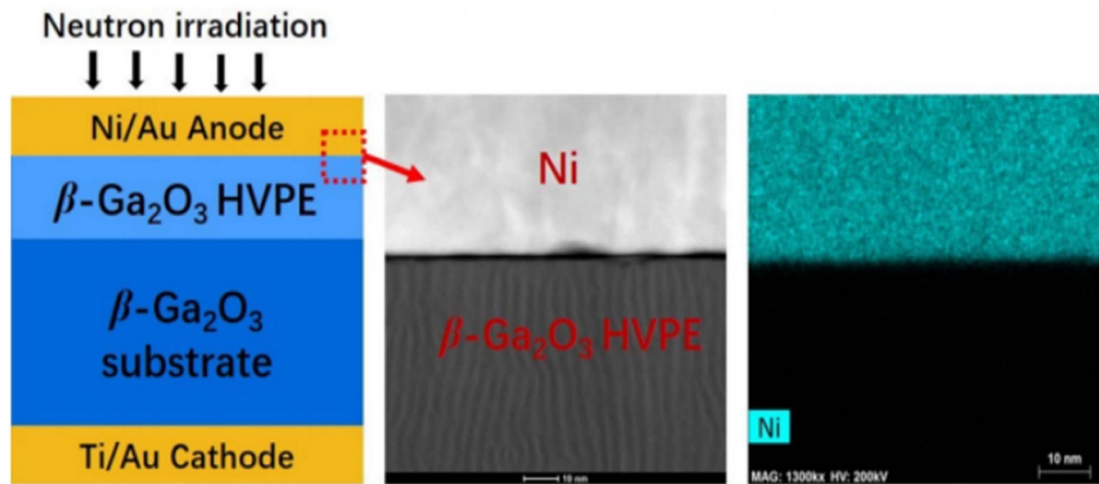


Figure 5. Schematic, TEM cross section of β -Ga₂O₃ SBDs and the distribution of Ni elements in the same area after irradiation by EDS tests. Reprinted from [35], with the permission of AIP publishing.

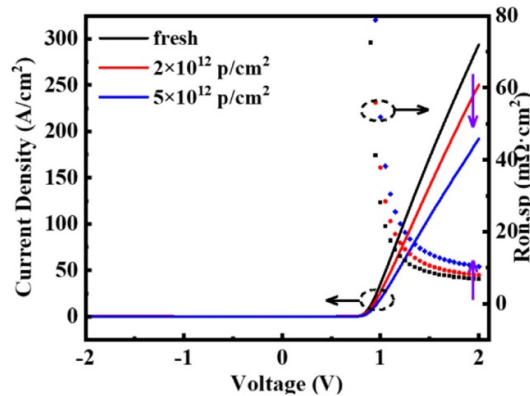


Figure 6. SRIM simulations of ion and damage profiles for 20 MeV proton incident into β -Ga₂O₃ SBDs. Reprinted from [39], with the permission of AIP publishing.

energies of the irradiation source exert varying effects on forward and reverse leakage currents. W. Fualso investigated the impact of injection dose on diode performance, revealing that electrical characteristics deteriorated significantly with increasing injection dose [36].

2.3.2. Proton irradiation

W. Du's group subjected gallium oxide Schottky diodes to 5 MeV proton irradiation. The resulting electrical characteristics resembled those observed after neutron irradiation: carrier concentration decreased, while the idealization factor and barrier height increased [37]. The concentration of deep-level defects (E_C -0.75 eV) exhibited a pronounced response to irradiation, indicating that non-interface states govern device degradation [38]. This aligns with the characteristic changes observed after neutron irradiation. Understanding the mechanism behind electrical parameter shifts is equally important as grasping their patterns. Analysis of reverse leakage current types indicates that reverse leakage is primarily driven by PFE. Y. Feng observed that the turn-on voltage of diodes increases after 20 MeV proton irradiation. As shown in figure 6 this occurs because lattice atoms shift during collisions, leaving vacancies at their original positions—forming Frenkel defects that act as recombination centers [39]. Following proton irradiation, the difference in breakdown voltage between SBDs (260 V–710 V) and RSBDs (440 V–790 V) decreased significantly, and the leakage current in both decreased; the difference in peak electric field between the two decreased, and the performance-enhancing effect of the NiO protective ring was suppressed [40].

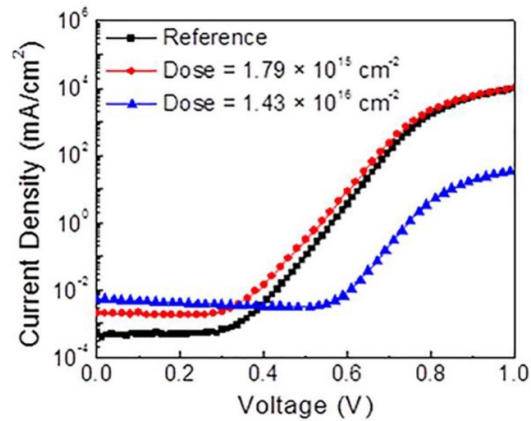


Figure 7. Forward current density voltage characteristics before and after electron irradiation with the fluence 1.79×10^{15} or $1.43 \times 10^{16} \text{ cm}^{-2}$. Reprinted from [41], with the permission of AIP publishing.

2.3.3. Electron irradiation

J Yang's group a diode fabricated on epitaxial $\beta\text{-Ga}_2\text{O}_3$ bulk material irradiated with a 1.5 MeV electron beam exhibited an 8% increase in the ideal factor at the highest injection level, with a corresponding reduction in reverse leakage current as shown in figure 7 [41]. The on resistance increased by more than two orders of magnitude. Following 1 MeV electron beam irradiation, the diode turn-on voltage decreased from 0.82 V to 0.77 V, the reverse current at -100 V bias decreased by 79%, and the Schottky barrier height decreased from 1.01 eV to 0.92 eV. These results indicate that electron irradiation at various doses leads to degradation of the electrical characteristics of $\beta\text{-Ga}_2\text{O}_3$ SBD [42]. To further elucidate the physical mechanism underlying electron irradiation induced degradation of diode electrical properties, Huang employed deep level transient spectroscopy to detect trap levels at 0.402 eV above the valence band edge, 0.626 eV and 0.755 eV below the conduction band edge [43]. Among these, the electron trap at 0.755 eV below the conduction band edge was identified as the primary defect state.

2.3.4. Heavy ion irradiation

Heavy ion irradiation can introduce defects such as Ga vacancies and O vacancies in $\beta\text{-Ga}_2\text{O}_3$ [44], forming amorphous damage tracks within the material. Beneath the Pt/PtO_x metal contact region, the substrate is susceptible to structural damage due to these damage tracks. These defects and damage tracks significantly degrade the electrical performance of $\beta\text{-Ga}_2\text{O}_3$ diodes, manifested by a six-order-of-magnitude increase in reverse leakage current density and an ideal factor rising from 1.07 to 1.38. Different heavy ion irradiation yield distinct effects. After 120 MeV Ag^{7+} irradiation as shown in figure 8 [15], the electrical characteristics of $\beta\text{-Ga}_2\text{O}_3$ Schottky diodes degrade due to the generation of numerous defect states resulting from increased oxygen defects and high-energy electron excitation post-irradiation. After 16 MeV 181Ta heavy ion irradiation, the barrier height decreased, leading to a drop in breakdown voltage [45]. This change pattern aligns with neutron irradiation results, indicating similar damage effects across different irradiation sources. Gao's group demonstrated that 212 MeV Ge ion irradiation can modulate defect energy levels and interface state distribution [46]; low-dose Ge ion irradiation optimizes the Au/Ni/ $\beta\text{-Ga}_2\text{O}_3$ interface quality, thereby enhancing the electrical performance of vertical-structure $\beta\text{-Ga}_2\text{O}_3$ SBD.

3. Control of defects in Ga_2O_3 power devices

The preceding sections have systematically summarized various defects present in $\beta\text{-Ga}_2\text{O}_3$ Schottky diodes during substrate growth and device fabrication. This chapter proposes corresponding improvement strategies for these defects: adopting novel growth techniques can suppress intrinsic defects during the material growth stage; pre-treating $\beta\text{-Ga}_2\text{O}_3$ substrates prior to device fabrication can further reduce internal substrate defects; For damage and defects introduced during the etching process of trench-type Schottky diodes, optimization of the etching process and post-etching treatment can be employed to mitigate issues; finally, methods for restoring the performance of GaO Schottky diodes under extreme environments such as irradiation are summarized.

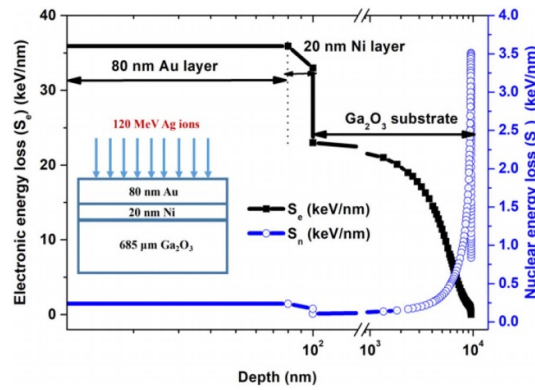


Figure 8. Electronic and nuclear energy losses of the 120 MeV Ag^{7+} ion beam as a function of depth in the Au/Ni/ $\beta\text{-Ga}_2\text{O}_3$ /Ti/Au SBDs; inset: Schematic of the Au/Ni/ $\beta\text{-Ga}_2\text{O}_3$ sample structure. Reprinted from [15], with the permission of AIP Publishing.

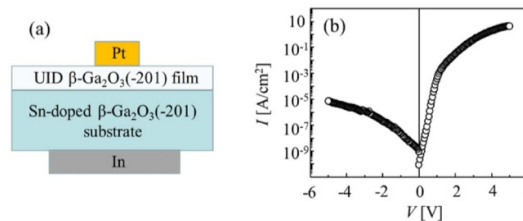


Figure 9. (a) Manufactured diodes by atmospheric pressure plasma-assisted CVD; (b) Schottky-like I–V characteristics of the SBD. Reproduced from [47]. CC BY 4.0.

3.1. New growth method

The performance of gallium oxide devices depends on the growth method of their epitaxial layers. Core evaluation metrics for epitaxial layer growth quality include crystal defect density, impurity content, thickness uniformity, surface flatness, and interface quality. These metrics influence three critical processes: ‘carrier transport,’ ‘electric field distribution,’ and ‘interface state density’ [48–53]. Therefore, the growth quality of epitaxial layers is crucial for gallium oxide devices.

Plasma-enhanced atomic layer deposition and plasma-enhanced chemical vapor deposition (PECVD) techniques have been used to grow $\beta\text{-Ga}_2\text{O}_3$ films at low temperatures between 60 °C–200 °C [54–56]. Crystallization requires post-deposition annealing at high temperatures (800 °C–1000 °C), which introduces unintended damage during fabrication. Consequently, Islam explored an efficient growth technique: atmospheric pressure PECVD [47]. This method enables $\beta\text{-Ga}_2\text{O}_3$ growth at 350 °C without high-temperature annealing. The effective impurity concentration is comparable to the unintentionally doped carrier concentration in epitaxial films grown using high-temperature techniques. The obtained barrier height is 2.2 eV, with the IV curve and cross-section shown in figure 9. Rebollo’s group proposed intentionally nitrogen-doped $\beta\text{-Ga}_2\text{O}_3$ films grown via plasma-assisted molecular beam epitaxy (PAMBE) in 2025 [57]. This method eliminates the need for ion implantation and avoids impurity migration and defect formation during high-temperature annealing. CV measurements (figure 10(b)) yielded a nitrogen doping range of 1.3×10^{18} – $4.5 \times 10^{19} \text{ cm}^{-3}$. This approach achieves high doping concentrations while minimizing the introduction of other impurities. It delivers extremely low series resistance, significantly reduces conduction losses, and enhances diode breakdown voltage and power density.

In 2025, He *et al* proposed a novel pulsed Al atom assisted *in situ* growth process, whose schematic principle is illustrated in figure 11 [58]. The red arrows represent the adsorption process of atoms, while the yellow arrows represent the diffusion process of atoms. Compared to films prepared using conventional methods, the $\beta\text{-Ga}_2\text{O}_3$ epitaxial layers grown by this process exhibit lower root-mean-square surface roughness and significantly reduced oxygen vacancy defect density. This effectively suppresses the random nucleation behavior of Ga atoms, thereby enhances the breakdown field strength of the material, as shown in figure 12.

Atmospheric pressure PECVD technology offers advantages such as low cost, large-area coverage, and high-volume production capabilities, making it well-suited for the mass fabrication requirements

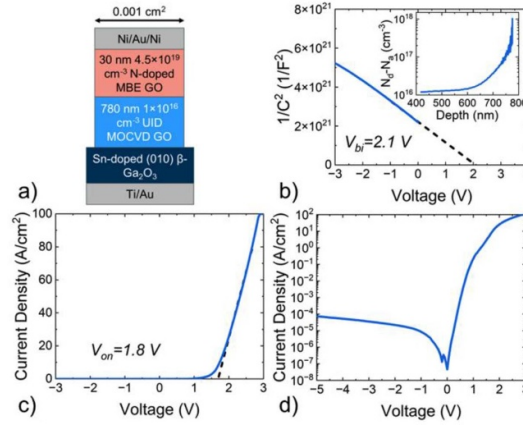


Figure 10. (a) Nitrogen-doped, n-junction device schematic. (b) $1/C^2$ vs voltage data at 1 MHz for the device in (a) along with the net charge vs doping (inset). J-V characteristics for the device in (a) using a (c) linear scale and (d) semi-log scale. Reproduced from [57], with the permission of AIP publishing.

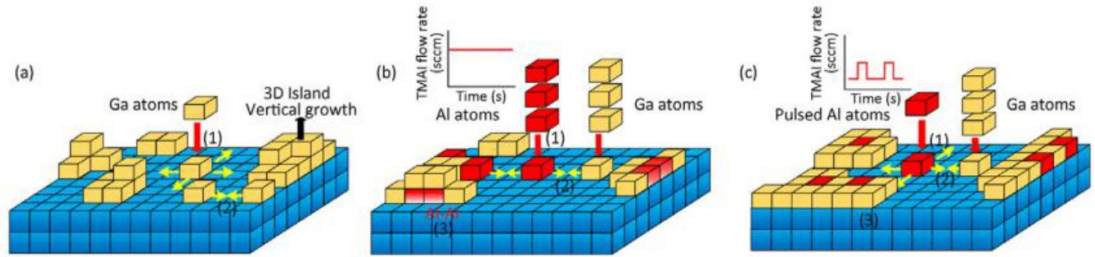


Figure 11. Schematic diagram of homoepitaxial growth mechanism on (001) β -Ga₂O₃ substrate under different growth conditions: (a) sample A, (b) sample B and sample C, (c) sample D. Reproduced from [58]. CC BY 4.0.

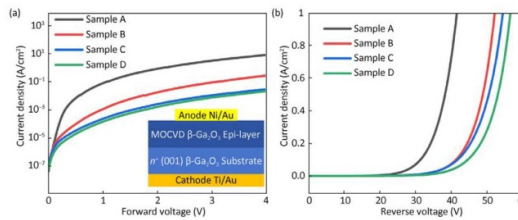


Figure 12. (a) Forward and (b) reverse I-V characteristics for the SBD. Reproduced from [58]. CC BY 4.0.

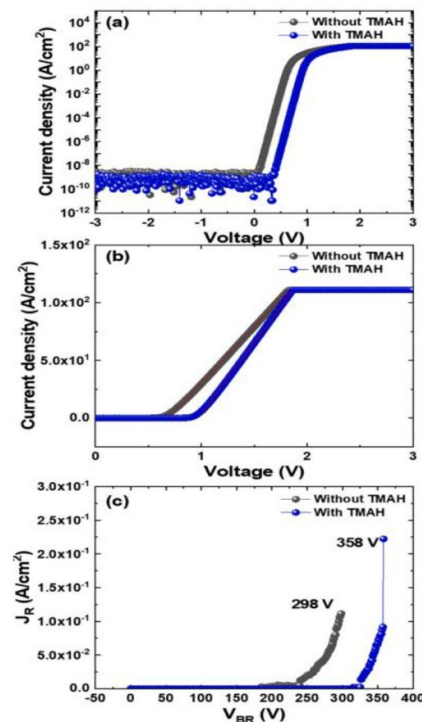
of large-size wafers. PBE for growing β -Ga₂O₃ films exhibit atomic-level precision and exceptional crystalline quality, making them ideal for high-precision prototype devices and fundamental research on ultrathin layers. Meanwhile, the novel pulsed Al atom-assisted *in-situ* growth technique demonstrates outstanding performance in low defect density, superior interface characteristics, and high-voltage operational stability. Among these, pulsed Al atom-assisted *in-situ* growth and PAMBE, leveraging their atomic-level film growth capabilities, can effectively suppress the aforementioned structural defects such as linear defects and microgrooves.

3.2. Preprocessing

The primary purpose of pretreatment is to eliminate surface contaminants, which mainly fall into two categories: hydrocarbons introduced from the environment, which readily form hydrophobic organic films on material surfaces; and inorganic particles and ions left over from the cleaning process. Additionally, during gallium oxide substrate processing or epitaxial layer growth, insufficient local oxygen partial pressure can induce the formation of oxygen vacancy defects. To address these issues,

Table 2. Effect of pre-treatment on the electrical characteristics of gallium oxide devices.

Preprocessing	Ideal factor	Barrier height (eV)	On resistance ($\text{m}\Omega \cdot \text{cm}^2$)	Switch-on voltage (V)	Breakdown voltage (V)	References
Tetramethylammonium hydroxide treatment	1.04–>1.02	0.99–>1.29	10.5–>8.2	0.56–>0.87	298–>398	[59]
2 min of HCl and 2 min of H_2O_2 washing	1.17–1.10	0.86–>0.92	10–>3.75	3.95–>1.65	423–>628	[60]
CF_4 plasma treatment	/	1.18–>1.31	4.6–>4.6	0.82–>0.95	150–>470	[61]
N_2 Oplasma treatment	/	0.92–>0.82	4.0–>3.5	0.8–>0.6	215–>323	[63]
O_2 plasma treatment	/	0.92–>1.2	4.0–>4.2	/	215–>318	[64]
O_2 Gas annealing treatment	1.09–>1.08	1.42–>1.45	/	/	931	[67]

**Figure 13.** (a) Semi-logarithmic room temperature J–V, (b) linear forward J–V and (c) reverse breakdown voltage curves of untreated and TMAH-treated Au/Ni/ β -Ga $_2$ O $_3$ Schottky diodes. Reprinted from [59], Copyright (2024), with permission from Elsevier.

extensive research has been conducted globally. Table 2 summarizes the effects of three surface treatment processes on the performance of Ga $_2$ O $_3$ Schottky diodes: solution cleaning, plasma treatment, and high-temperature annealing in an atmosphere.

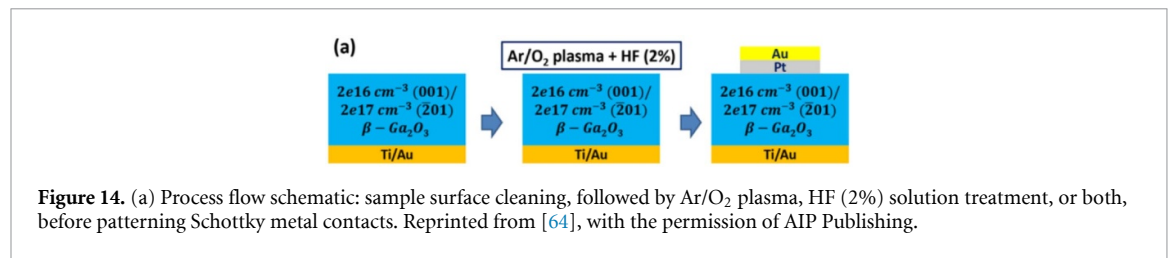
3.2.1. Solution processing

Janardhanam's *et al* Treatment of β -Ga $_2$ O $_3$ SBD with tetramethylammonium hydroxide effectively reduced the surface roughness of β -Ga $_2$ O $_3$ and efficiently removed surface oxides and contaminants formed by chemical adsorption [59]. This resulted in a significant improvement in their electrical properties, as shown in figure 13.

When chemical cleaning using HCl, BOE, or HF is the mainstream traditional method, the drawback is the requirement for prolonged exposure. Jeon's group proposed an innovative cleaning method that avoids prolonged exposure damaging the surface quality of gallium oxide. Employing a cleaning sequence of 2 min with HCl followed by 2 min with H_2O_2 minimizes surface roughness [60]. This approach also demonstrates lower leakage current, higher breakdown voltage, and reduced on-resistance.

3.2.2. Plasma treatment

Luo *et al* achieved a four-order reduction in reverse leakage current for β -Ga $_2$ O $_3$ SBD treated with low-power CF_4 plasma, without compromising forward characteristics [61]. Ji's group employed N plasma



surface treatment to achieve N-type dopant incorporation on Ga₂O₃ films [62]. The N-doped β -Ga₂O₃ films demonstrated excellent defect compensation for oxygen vacancy defects, thereby enhancing the electrical performance of Ga₂O₃ based surface devices. In the same year, He is group investigated two oxygen-containing plasma-treated β -Ga₂O₃ SBD: N₂O plasma treatment and O₂ plasma treatment [63]. The oxygen-containing plasma partially reduced the deep-level trap density in the anode region, effectively improving the device surface quality. Sharma *et al* employed Ar/O₂ plasma exposure followed by HF treatment (as shown in figure 14). This process yielded interfaces with low interface state density, increased barrier height by 0.3 eV, and elevated breakdown voltage (absolute value of 440 V) by over 100 V [64]. Plasma pretreatment fundamentally compensates for surface oxygen vacancies, thereby improving β -Ga₂O₃ interface quality. Ultimately, this approach reduces oxygen vacancy defects on the GaO surface, along with particulate impurities and organic contamination from the atmosphere.

3.2.3. High-temperature gas annealing

The primary function of high-temperature annealing for both inherent process defects and irradiation-induced defects in diodes is to repair unstable defects, activate doping, reduce deeps, and restore crystal integrity.

Anomalous reverse leakage characteristics were observed in β -Ga₂O₃ SBD grown by halide vapor phase epitaxy, exhibiting a high density of oxygen vacancy states. This thin surface layer permits electron tunneling, leading to the anomalous reverse leakage behavior. Lingaparathi annealed β -Ga₂O₃ epitaxial layers in an oxidizing atmosphere to passivate surface oxygen vacancy states [65], significantly reducing reverse leakage current. Lingaparathi annealed β -Ga₂O₃ epitaxial layers for 40 min in an oxygen atmosphere [66]. Oxygen annealing proved surface states and reducing net carrier concentration, thereby modulating the reverse leakage of β -Ga₂O₃. By optimizing oxygen annealing to compensate for oxygen vacancy-related surface states, reverse leakage characteristics were reduced by two orders of magnitude. Hu employed low-temperature O₂ gas annealing to surface-treat β -Ga₂O₃ SBD for enhanced electrical performance [67]. A 5 min O₂ gas treatment at 400 °C on the SBD surface resulted in oxidation of surface Ga²⁺ states, as revealed by x-ray photoelectron spectroscopy analysis. Compared to control devices, this treatment increased the breakdown voltage by 18% and improved the power figure of merit by 42%.

Solution pretreatment mainly addresses low temperature, low cost, surface cleaning and light etching only, does not change the bulk defects, and cannot deeply passivate oxygen vacancies, while plasma pretreatment can accurately modulate the surface defects and chemical state O filling, but there is a risk of plasma bombardment damage. High-temperature gas pretreatment provides high temperature, deep passivation, repair of lattice and oxygen vacancies, and significant reduction of interfacial states, but has a high thermal budget and may lead to surface roughening.

3.3. New etching method

In 2021, Tang's group proposed a novel β -Ga₂O₃ etching process. β -Ga₂O₃ trench SBD fabricated using a self reactive etching (SRE) process—involving a mixed BCl₃ and Cl₂ gas via inductively coupled plasma reactive ion etching (ICP-RIE)—exhibited excellent surface quality and superior electrical performance [68]. Due to the unique chemical reactions in SRE, no impurities or ion-induced damage were introduced, and the surface was repaired and reconstructed. Deng demonstrated that modifying the BCl₃: Ar ratio in the etching plasma alters the surface quality of β -Ga₂O₃ SBD [69]. Improved surface quality enhances electrical performance.

Oshima *et al* introduced a novel plasma free anisotropic etching technique for β -Ga₂O₃ substrates [70]. This technique employs a simple and safe gas etching process using a synthetic gas mixture (H₂/N₂ containing 3.96 vol% H₂) at atmospheric pressure. This method avoids plasma-induced damage and produces non-toxic gases. Due to nanoscale removal, the Schottky barrier potential can be modulated by controlling the intermediate layer, enabling precise regulation of the turn-on voltage. A significant drawback is the requirement for high-temperature processing.

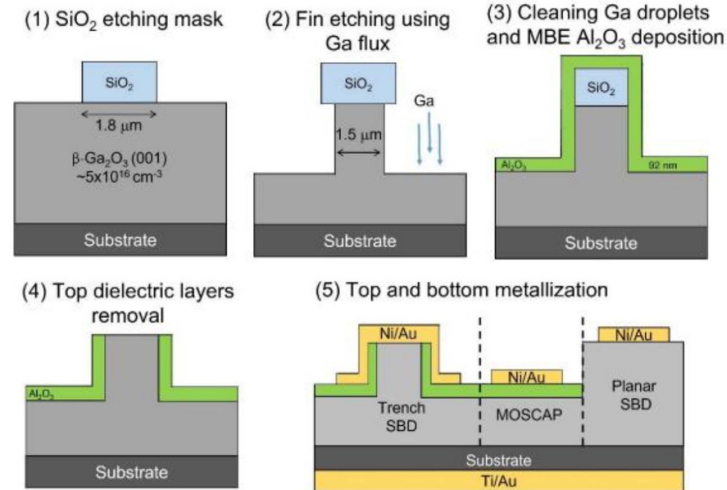


Figure 15. Process flow for the Ga-flux etched trench SBD. Reprinted from [71], with the permission of AIP publishing.

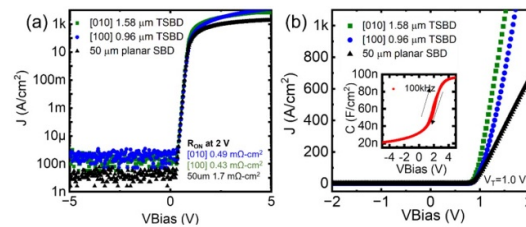


Figure 16. Forward current-voltage characteristics of the planar and trench SBDs (TSBD) in (a) log and (b) linear scale with capacitance voltage characteristics of the MOS capacitor on the etched surface in the inset. Reprinted from [71], with the permission of AIP publishing.

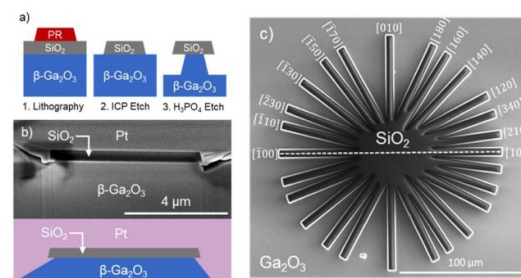


Figure 17. (a) A schematic of the etch process. (b) FIB SEM micrograph and diagram of a [160]-oriented spoke cross section. (c) SEM micrograph and diagram of a wagon wheel structure after etching. The dashed line indicates the mirror plane. Reprinted from [73], with the permission of AIP publishing.

S. Dhara's group proposed fabricating β -Ga₂O₃ trench SBD using gallium atomic beam etching technology, as shown in figure 15 [71]. These diodes exhibit excellent field strength and power device figure-of-merit, as illustrated in figure 16. This method is suitable for producing trench Schottky diodes with relatively complex structures, and when combined with SEM, it enables the creation of desired structures. However, the drawbacks are also evident: slow etching rates and a cumbersome process.

Zhang and Rebollo employed heated phosphoric acid etching [72, 73]. Following wet etching in a 140 °C phosphoric acid solution, the ideal coefficient of the sidewall Schottky diode approached 1 with no hysteresis phenomenon. This etching method achieved etch rates comparable to those obtained with dry etching techniques, overcoming the drawback of slow wet etching rates. The etched (001) surface exhibits a smoother surface than that of an inductively coupled plasma-etched surface, with all spokes featuring relatively smooth sidewalls as shown in figure 17. This etching method can serve as an alternative to dry etching.

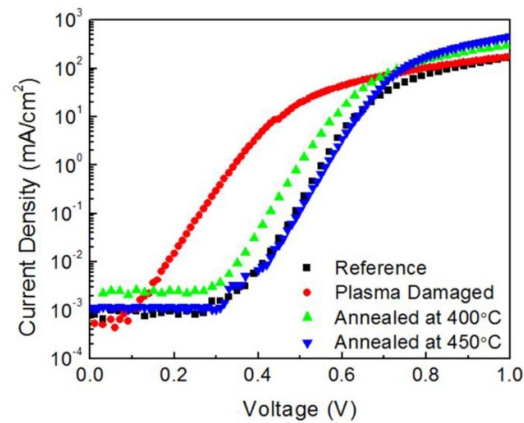


Figure 18. Forward I–V characteristics from Ni/Au Ga₂O₃ diodes on surfaces dry etched in BCl₃/Ar ICP discharges and annealed at 400 or 450 °C prior to metallization. Reprinted from [16], with the permission of AIP publishing.

However, it is prone to splatter and burns during operation, necessitating strict protective measures and ventilation obtained with dry etching techniques, overcoming the drawback of slow wet etching rates. The etched (001) surface exhibits a smoother surface than that of an inductively coupled plasma-etched surface, with all spokes featuring relatively smooth sidewalls as shown in figure 17. This etching method can serve as an alternative to dry etching. However, it is prone to splatter and burns during operation, necessitating strict protective measures and ventilation.

3.4. Damage control following etching

During trench diode fabrication, defects such as lattice damage, amorphous layers, and surface contamination form on the trench sidewalls. These defects create additional recombination centers and leakage channels, leading to increased leakage current and degraded breakdown characteristics in the device. Effective remediation requires subsequent cleaning, annealing, or sidewall passivation processes.

As mentioned earlier, BCl₃ plasma etching is an effective method for forming trenches in β -Ga₂O₃, offering high etch rates. However, the resulting damage can degrade device performance. A. R. Gutierrez's group employed post-processing methods such as mild (80 °C) H₃PO₄, (70 °C) tetramethylammonium hydroxide [74], and room-temperature 10:1 dilute hydrochloric acid BCl₃ plasma etching. This yielded diodes with high breakdown voltage and low on-resistance, measuring 1680 V and 4.78 mΩ · cm², respectively. The comparison of electrical measurements is shown in figure 17. Similarly, Lee *et al* performed post-processing of dry-etched β -Ga₂O₃ using scanning electron microscopy or TMAH solutions (with sulfuric acid and hydrogen peroxide mixture or tetramethylammonium hydroxide), effectively restoring damage to β -Ga₂O₃ caused by dry etching without leaving any residues on the surface [27]. Yang's group discovered that thermal treatment at 450 °C could eliminate near-surface damage induced by the dry etching process of β -Ga₂O₃ [16]. This treatment is sufficient to essentially fully restore the forward and reverse I–V characteristics of the diodes fabricated on these surfaces, as shown in figure 18. Post-annealing enabled JTE structures to fully optimize the electric field distribution. This resulted in higher breakdown voltages and more stable forward conduction characteristics.

3.5. Defect control in irradiation

Compared to other semiconductor materials, gallium oxide inherently possesses a certain degree of intrinsic radiation resistance [76]. Adopting countertop terminal structures and field plate structures [77] can effectively enhance the device's resistance to particle irradiation while significantly increasing the breakdown voltage. Beyond structural optimization, hydrogenation treatment can also strengthen GaO's radiation resistance [78]; hydrogenation introduces a large number of oxygen vacancy defects into GaO, thereby significantly improving the material's radiation stability.

The preceding section addressed enhancing the radiation resistance of gallium oxide Schottky diodes prior to irradiation. Next is damage recovery after irradiation. Lin *et al* [75] discovered that defects induced by proton irradiation can be mitigated through 500 °C annealing heat treatment, as shown in figure 19. Here, P denotes the original sample, A denotes the annealed sample, and D1 and D2 denote irradiated samples. Comparing D2-P and D2-A500 illustrates this effect. Liao's group restored the lattice

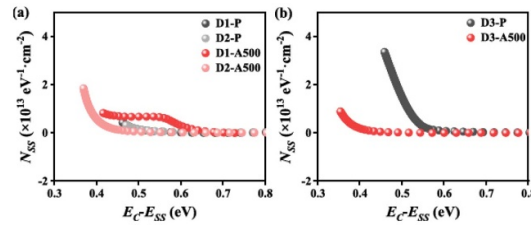


Figure 19. The distribution of $N_{SS} \sim E_C - E_{SS}$ for (a) irradiated and (b) unirradiated devices. Reprinted from [75], with the permission of AIP publishing.

of epitaxial β -Ga₂O₃ layers irradiated with 212 MeV Ge ions through high-temperature (500 °C) nitrogen annealing [79]. This demonstrates that high-temperature annealing is an effective method for restoring device electrical properties following irradiation by heavy ions, protons, or other types of radiation. Azarov *et al* irradiated β -Ga₂O₃ epitaxial layers with 400 keV $^{58}\text{Ni}^+$ ions at a dose of $2 \times 10^{14} \text{ cm}^{-2}$ during irradiation [79], followed by high-temperature annealing at 300 °C. The observed significant reduction in lattice disorder within the β -Ga₂O₃ indicates that high-temperature implantation can effectively suppress the formation of radiation-induced defects. Hu *et al* optimized the conditions of the high-temperature implantation process to enhance impurity activation [80], thereby further mitigating irradiation-induced damage.

4. Conclusions

Gallium oxide offers potential advantages for high-power devices due to its UWBG and high breakdown voltage, particularly in applications requiring high-voltage support such as electric vehicles, industrial power systems, and power transmission. However, despite β -Ga₂O₃'s excellent properties, challenges remain in optimizing its performance. Owing to the inherent advantages of β -Ga₂O₃ material, it holds significant potential for development in power devices. However, intrinsic defects in β -Ga₂O₃ and limitations in its lattice structure make it difficult to form p-type doping, thereby restricting the advancement of bipolar devices. Consequently, unipolar devices have seen rapid development. Consequently, unipolar devices have seen rapid development. This review summarizes the defects inherent to β -Ga₂O₃ Schottky diodes themselves and those arising from the fabrication process, along with methods for controlling these defects. Additionally, process-induced damage during β -Ga₂O₃ Schottky diode fabrication is addressed. Numerous approaches exist to mitigate defects in β -Ga₂O₃ Schottky diodes, including surface pretreatment of β -Ga₂O₃, novel epitaxial growth techniques, advanced etching methods, and post-etching treatments.

Future advancements will offer improved methods for controlling defects in gallium oxide Schottky diodes. While these techniques may already be applied to other devices, they can also be adapted for use in gallium oxide Schottky diodes. It is hoped this paper will contribute to the advancement of β -Ga₂O₃ Schottky diodes in the future.

Data availability statement

The data cannot be made publicly available upon publication because no suitable repository exists for hosting data in this field of study. The data that support the findings of this study are available upon reasonable request from the authors.

Author contributions

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